Sliding, swinging and rolling

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose the energy terms that match the type of motion.

SUCCESS CRITERIA - I can...

1. Distinguish translational and rotational kinetic energy.
2. Use equal height changes to compare ideal energy changes.
3. State when a simple point-mass model is insufficient.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 4 kg object descends 6 m near Earth. Use g=10. Find ΔU and the work done by gravity.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K_{translation} = \frac{1}{2}mv^2$; $K_{total} = K_{translation} + K_{rotation}$. A rolling object also rotates. Track all energy forms; an ideal pendulum exchanges gravitational and kinetic energy.

Worked reasoning: An object loses 20 J of gravitational potential energy. If 8 J becomes rotational kinetic energy and losses are negligible, only 12 J becomes translational kinetic energy.

Your turn: Draw three energy bars and explain why assigning all 20 J to forward speed would be wrong.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A slider and a rolling wheel start at the same height from rest. Why can the wheel reach the bottom with a lower center speed?

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

04 Apply the physics

A solid cylinder rolls without slipping from height 1.5 m. Use $I=\frac{1}{2}mr^2$ and g=10. Derive v from $mgh=\frac{1}{2}mv^2+\frac{1}{2}I(v/r)^2$. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Sliding, swinging and rolling

antocciphysics

Continue down each column.

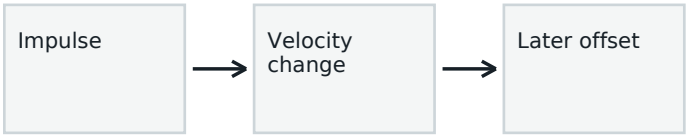
05 Explain your reasoning

A pendulum’s tension is perpendicular to its instantaneous motion in the ideal fixed-length model. What work does tension do, and which interaction changes K?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep impulse fixed, multiply target mass by 4 and lead time by 6. Compare the modeled transverse shift using an equation.
Label the quantities in the model and state its assumptions.



Marzano check - circle: 1 2 3

07 Exit check

A point-mass pendulum drops 0.80 m vertically from rest. Find speed at bottom using g=10 and neglect drag. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A thin hoop rolls without slipping from rest down the same 1.5 m height as the solid cylinder in question 4. Use $I = mr^2$ and $g = 10\ m/s^2$. Derive its final center speed and compare with the cylinder.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3